aircraft, but the scope is incredibly vast. Drones, by definition, are aircraft capable of operating without a pilot on board. This means that from tiny toy drones to large drones that can ferry humans, they all fall under the same umbrella. One might associate drones with photography, surveillance, or agriculture. However, the applications are limitless. Hollywood has shown us drones that chase cars in high-octane sequences, but did you know the world is inching towards drone taxis? Venturing beyond India's boundaries, the drone narrative becomes even more captivating. US-based firms have successfully dispatched over 400,000 critical medicines and vaccines in Africa using drones. Giants like Google are now leveraging drones for e-commerce parcel deliveries.

But here's the kicker—drone taxis aren't a distant dream. Chinese tech mogul EHang has already crafted air taxis. These aren't mere prototypes but production-ready vehicles. Just punch in your location, hop on, and off you go on a thrilling, autonomous aerial adventure. And if that doesn't pique your curiosity, NASA's drone flight on Mars surely will. The drone era isn't confined to our planet; it's venturing into the vast cosmic frontiers.

Drones are more than just buzzing devices. They're a testament to human innovation, bridging gaps and creating industry opportunities. As the world soars into this new age of drone technology, India, with its burgeoning drone ecosystem, is poised to be a significant player in this aerial symphony. The sky is merely the beginning. The drone's story isn't just about fancy gadgets zipping across the sky; it's about a technological revolution radically transforming India's defence, agriculture, healthcare, and countless other sectors.

Drone propulsion: Why not ducted fans?

Despite diverse applications, most drones use open propellers for propulsion. This raises the question: Why aren't alternatives like ducted fans, similar to jet engines, more prevalent, especially considering the risks of open propellers? Let's explore the core reasons.

Safety and efficiency. Safety and efficiency are vital in drone technology. While ducted fans may seem safer, they are only sometimes the most efficient, as evidenced by their absence in helicopters. The commercial drone industry has robust safety features to address failures, allowing recovery or shutting down the drone to prevent ongoing risks.

Exploring new designs. Innovation remains open-ended. While open-propeller drones are prevalent, alternative designs, like electric ducted fans with potential power lift benefits, are being explored. Any new technology needs thorough testing before broad adoption. Innovators need to collaborate with top companies to validate new design efficiencies.

In conclusion, while the drone industry predominantly features open-propeller drones, ducted fans still hold potential, especially for high-speed or safety-critical scenarios, and it's not for lack of consideration for other technologies. Balancing efficiency with safety is crucial. As technology advances, overcoming challenges like efficiency and weight becomes possible. Encouraging innovation and collaboration could see ducted fans play a more significant role in future drone developments.

An innovation transforming industries

The first image that comes to mind might be colossal drones—almost mirroring fighter jets, with an impressive range extending over 500 kilometres. They soar at altitudes rivalling commercial jets. But drones aren't limited to military use. The range of applications is vast, from micro-surveillance drones that aid soldiers in reconnaissance missions to sophisticated drones that can launch missile-like munitions. Yet, move past the defence realm, and drones take on new roles. For instance, imagine the scenario in agriculture where drone technology replaces traditional pesticide spraying. Instead of manual labourers toiling in fields, exposing themselves to harmful chemicals, a drone achieves the same in a fraction of the time, with considerable efficiency, and without human exposure. They have numerous applications:

Healthcare. Delivering essential medicines and vaccines to remote areas.

E-commerce. Companies like Google and Bing are exploring drone deliveries.

Transportation. Drone taxis, developed by companies like EHang, are becoming a reality. By the way, they have built more than 30 such air taxis.

Defence. Drones, varying in size from toy-like to fighter jet proportions, offer from surveillance to offensive capabilities. Some can cover 500km without refuelling, flying as high as commercial airlines. While applicable in military and disaster situations, they also have high-end applications in healthcare and agriculture. Some, resembling missile launchers, deploy up to 50 drones to target and destroy locations. Their price, often over three million rupees, suggests a lucrative market.

Agriculture. Drones are utilised for crop health assessment and pesticide and fertiliser application, and in environmental and forestry sectors for wildlife monitoring, forest conservation, mapping, and designing water catchment areas. In agriculture, labourers traditionally spray pesticides using 200 litres of water for an acre, taking two hours and risking chemical exposure. Drones streamline this, using only 10 litres of water and less pesticide